

MALAVIKA MADHAVAN

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SUMMARY

Computer Science Engineering student at SRM Institute of Science and Technology with strong foundations in Data Structures, Algorithms, Object-Oriented Programming, and Python. Solved 500+ DSA problems across coding platforms and developed projects using Python, OpenCV, and Machine Learning. Seeking internship opportunities to apply technical skills, gain industry experience, and contribute to real-world problem solving.

S.R.M Institute of Science and Technology

August 2024 - Present

Bachelor of Technology in Computer Science and Engineering - CGPA - 9.48 (till date)

TECHNICAL SKILLS

- Languages: C, C++, Python, JavaScript, HTML, CSS
- Libraries & Frameworks: OpenCV, NumPy, Pandas, MediaPipe
- Tools: VS Code, Git, GitHub
- Core Subjects: Data Structures & Algorithms, OOP, Operating Systems, SQL

PROJECTS

AI-Powered Virtual Mouse Control | [GitHub](#)

Jan 2026

Tech Stack: Python, OpenCV, MediaPipe, PyAutoGUI

- Built a touch-less virtual mouse using computer vision to control cursor movement through hand gestures.
- Implemented real-time tracking of 21 hand landmarks using MediaPipe for cursor navigation and click detection via pinch gesture
- Applied coordinate scaling and image mirroring to ensure smooth and accurate cursor control across different screen resolutions.

Motion-Activated Security Alarm System | [GitHub](#)

Dec 2025

Tech Stack: Python, OpenCV, Winsound

- Developed a **motion detection security system** using frame differencing and contour analysis to continuously monitor video frames and identify environmental changes in real time.
- Applied **image preprocessing techniques** including grayscale conversion, Gaussian blurring, and binary thresholding to reduce noise and improve the accuracy of motion detection.
- Implemented a **trigger mechanism based on contour area analysis** to detect significant movement and automatically activate an audible system alarm when a potential security breach is identified.

Handwritten Digit Recognition System | [GitHub](#)

Sept 2025

Tech Stack: Python, TensorFlow/Keras, NumPy.

- Developed a machine learning model to recognize handwritten digits (0-9) using image classification techniques with 95% accuracy.
- Trained the model on the MNIST dataset and implemented preprocessing methods such as normalization and re-sizing.
- Achieved accurate digit prediction by building and evaluating a neural network-based classifier.

EXTRA CURRICULAR

techvayuna club
design team

Oct 2024 - Present

- Managed and coordinated 404 Showdown, a technical event focused on testing participants' basic programming and problem-solving skills.

Campus life - Marketing, Content

editing team - I have been part of campus life since 1st year 2nd semester. I have collaborated to do a bunch of reels, edit the same and have taken leadership role when I had to step in.

ACHIEVEMENTS

- CodeChef: 2* Rated Competitive Programmer. (1510)
- LeetCode: Solved 400+ problems; Codeforces: Rated Pupil in competitive programming contests.